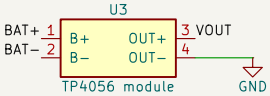
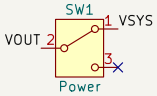
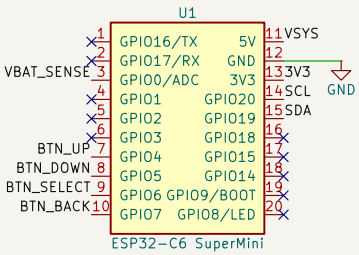
UtilWatch perfboard demo

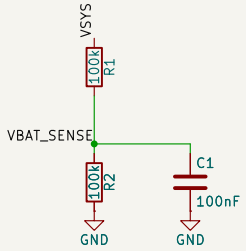
Same wiring as the working firmware: OLED on GPIO19 (SDA) / GPIO20 (SCL),
buttons on GPIO4–7 to GND with the internal pull-ups.

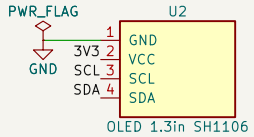


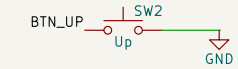


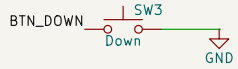


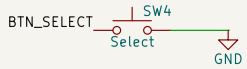


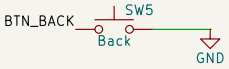












Header order is from mischianti.org's SuperMini pinout:
check it against the silkscreen on your board.

Battery, NOT TESTED YET:
- Charge through the TP4056's USB-C. GND is the TP4056 OUT−, not B− (B− sits behind its protection switch).
- VSYS goes into the SuperMini 5V pin and its own 3.3 V regulator. Below about 3.5 V on the cell the 3V3 rail starts to sag.
- Turn SW1 off before plugging USB into the SuperMini, or its USB 5 V pushes into the battery.
- R1/R2 halve VSYS for GPIO0 (ADC1_CH0): 4.2 V reads as 2.1 V. They draw about 20 uA while SW1 is on.

Perfboard demo before a real PCB

Sheet: /
File: utilwatch.kicad_sch

Title: UtilWatch

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 1/1